

Theme showcase

PSYC 51.07: Models of language and conversation

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Simple bullet points

- First main point
- Second main point
- Third main point

Text emphasis

- Use **bold** for strong emphasis
- Use *italics* for subtle emphasis
- Use `code` for inline code

Code example

```
1  def hello():  
2      """Simple function example."""  
3      print("Hello, Dartmouth!")  
4      return True
```

Long code (auto-split demo)

```
1 class MultiHeadAttention(nn.Module):
2     """Multi-head attention mechanism for transformers."""
3
4     def __init__(self, d_model, num_heads, dropout=0.1):
5         super().__init__()
6         self.num_heads = num_heads
7         self.d_k = d_model // num_heads
8
9         self.W_q = nn.Linear(d_model, d_model)
10        self.W_k = nn.Linear(d_model, d_model)
11        self.W_v = nn.Linear(d_model, d_model)
12        self.W_o = nn.Linear(d_model, d_model)
13        self.dropout = nn.Dropout(dropout)
14
15    def forward(self, query, key, value, mask=None):
16        batch_size = query.size(0)
17
18        # Linear projections
19        Q = self.W_q(query)
20        K = self.W_k(key)
```

continued...

Long code (auto-split demo)

```
21     V = self.W_v(value)
22
23     # Reshape for multi-head attention
24     Q = Q.view(batch_size, -1, self.num_heads, self.d_k)
25     K = K.view(batch_size, -1, self.num_heads, self.d_k)
26     V = V.view(batch_size, -1, self.num_heads, self.d_k)
27
28     # Compute attention scores
29     scores = torch.matmul(Q, K.transpose(-2, -1))
30     scores = scores / math.sqrt(self.d_k)
31
32     if mask is not None:
33         scores = scores.masked_fill(mask == 0, -1e9)
34
35     attention = F.softmax(scores, dim=-1)
36     attention = self.dropout(attention)
37
38     # Apply attention to values
39     output = torch.matmul(attention, V)
40     output = output.view(batch_size, -1, self.d_model)
```

...continued...

Long code (auto-split demo)

```
41 |  
42 | return self.W_o(output)
```

...continued

Equation

The attention mechanism:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

Interleaved text and equations

The softmax function normalizes scores into probabilities:

$$\text{softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

We scale by the square root of dimension to prevent vanishing gradients:

$$\text{score} = \frac{QK^T}{\sqrt{d_k}}$$

This keeps variance stable during training.

Equations in bullet lists

- The dot product measures similarity between vectors
- Attention scores: $\text{score}(q, k) = q \cdot k$
- Scaled version prevents large values:

$$\text{score}(q, k) = \frac{q \cdot k}{\sqrt{d_k}}$$

- Output is weighted sum of values

Equations in numbered lists

1. Compute query, key, value projections
2. Calculate attention scores:

$$A = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right)$$

3. Apply attention to values: $\text{output} = AV$
4. Project back to model dimension

Simple table

Model	Year	Params
GPT-2	2019	1.5B
GPT-3	2020	175B
GPT-4	2023	Unknown

GPT-3	Decoder-only transformer	Common Crawl, books, Wikipedia	Few-shot learning, text generation, code completion
T5	Encoder-decoder transformer	C4 (Colossal Clean Crawled Corpus)	Text-to-text framework, translation, summarization
			Open

Long table (auto-split demo)

Model	Architecture	Training data	Key capabilities
BERT	Encoder-only transformer	BooksCorpus, Wikipedia	Bidirectional context, masked LM
GPT-2	Decoder-only transformer	WebText	Text generation, zero-shot
GPT-3	Decoder-only transformer	Common Crawl, books	Few-shot learning, code completion
T5	Encoder-decoder transformer	C4 Corpus	Text-to-text, translation
XLNet	Transformer-XL	BooksCorpus, Wikipedia	Permutation LM, long context
RoBERTa	Encoder-only transformer	Extended pretraining	Robust optimization of BERT
ALBERT	Encoder-only transformer	Same as BERT	Parameter sharing, efficient
ELECTRA	Encoder-only transformer	Same as BERT	Replaced token detection

continued...

Long table (auto-split demo)

Model	Architecture	Training data	Key capabilities
DeBERTa	Encoder-only transformer	Wikipedia, books	Disentangled attention
LLaMA	Decoder-only transformer	Public data only	Open weights, efficient
LLaMA 2	Decoder-only transformer	2T tokens	RLHF, chat-optimized
Mistral	Decoder-only transformer	Undisclosed	Sliding window attention
Claude	Decoder-only transformer	Web, books, code	Constitutional AI
Claude 2	Decoder-only transformer	Extended training	Longer context, improved
GPT-4	Decoder-only transformer	Undisclosed	Multimodal, reasoning
Gemini	Decoder-only transformer	Multimodal data	Native multimodal

...continued...

Long table (auto-split demo)

Model	Architecture	Training data	Key capabilities
PaLM	Decoder-only transformer	780B tokens	Pathways, chain-of-thought
Falcon	Decoder-only transformer	RefinedWeb	Open source, efficient

...continued

Two-column layout

Left column

- Point A
- Point B

Right column

- Point X
- Point Y

Quote

//

*This line is a quote that demonstrates the
blockquote styling from the CDL theme.*

//

— Anonymous

content.

Example

Consider the function $f(x) = x^2$. When $x = 3$, we get $f(3) = 9$.

Warning

Training large language models requires

Note

The attention mechanism was first introduced in the context of machine translation (Bahdanau et al., 2014), allowing models to focus on relevant parts of the input sequence. Key insight: Instead of compressing the entire input into a fixed-length vector, attention allows the decoder to "look back" at the encoder states.

Example

Computing self-attention for a sequence:

- Input: "The cat sat"
- Query for "cat": looks for related words
- High attention to "The" (determiner) and "sat" (verb)
- Low attention to unrelated tokens

Warning

Common pitfalls when fine-tuning LLMs:

- Overfitting on small datasets
- Catastrophic forgetting of pre-trained knowledge
- Learning rate too high can destabilize training
- Always validate on held-out data!

Multiple outputs

Note

Transformers process all tokens in parallel, unlike RNNs which process sequentially.

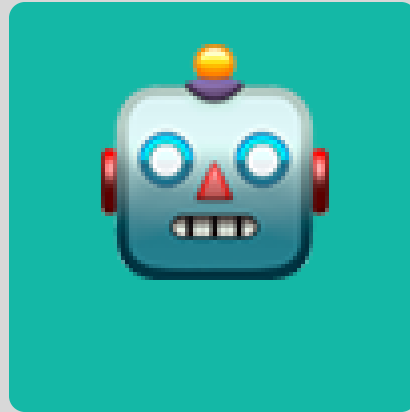
Warning

The quadratic memory complexity of attention limits context length.

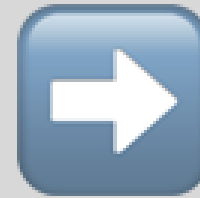
Emoji figures



Students



LLM



Insights

Emoji sizes and states



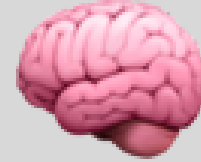
xs
(24px)



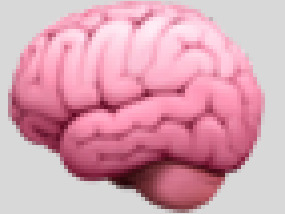
sm
(48px)



md
(72px)



lg
(96px)



xl
(128px)



Normal

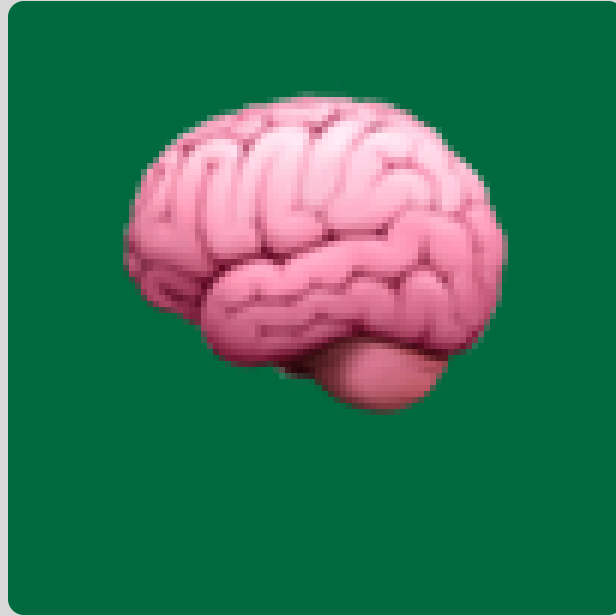


Faded



Inactive

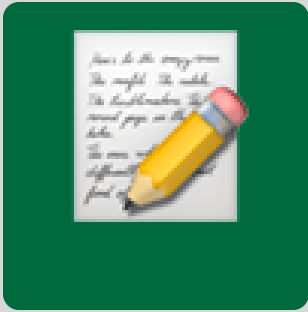
Emoji xxl (hero) size



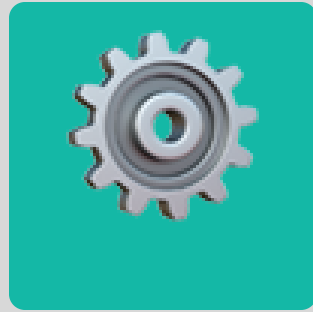
xxl (192px) - Hero size

Perfect for title slides or featuring a single concept prominently.

Emoji grid layout



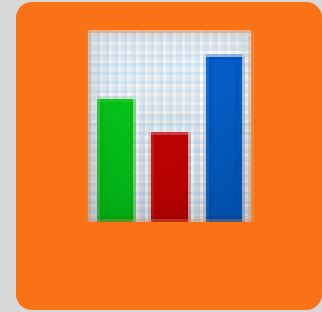
Input



Process



Analyze



Output

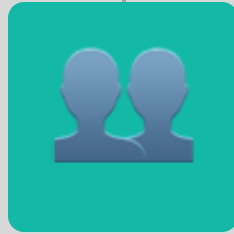
Emoji hierarchy structure



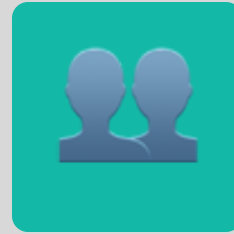
Organization



Team A



Team B

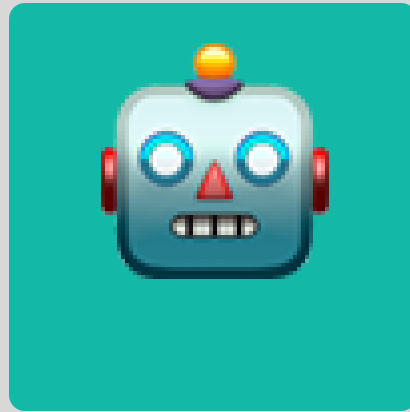


Team C

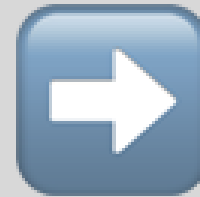
Labeled emoji example



Training
data



Languag
e model



Generat
ed text

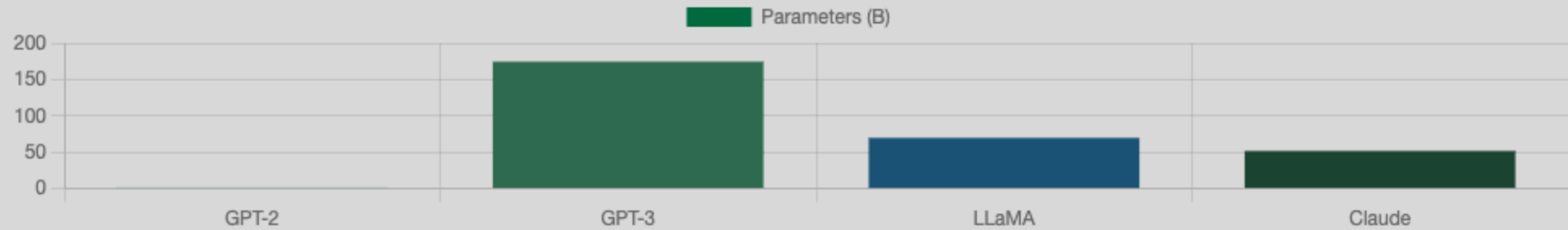
Diagram container example

Error 400: Unable to decode the source.

A simple flowchart rendered via Kroki.io

Note: Diagrams are pre-rendered SVGs using the Kroki service.
Supports Mermaid, GraphViz, PlantUML, and more.

Chart container example



Model parameter comparison (Chart.js)

Next steps

- Review: Chapter 3 of *Speech and Language Processing*
- Explore: Hugging Face Transformers documentation
- Try: Build a simple chatbot with the provided templates

Assignments

- Problem Set 2 due next Friday
- Final project proposal due in 2 weeks

Course resources

- Canvas: Course materials and submissions
- GitHub: Code examples and notebooks
- Office hours: Tuesdays 2-4pm, Moore 354

Contact

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